Calculus II

Problems for Training #4 10–14 Mar 2025

Topics:

- Gradient vectors
- Tangent planes and differentials
- Linear approximations
- Directional derivatives

Problem 1. Find the gradient of the function at the given point. Then sketch the gradient together with the level curve that passes through the point.

$$g(x, y) = xy^2, \quad (2, -1).$$

Problem 2. Find ∇f at the given point.

$$f(x, y, z) = x^2 + y^2 - 2z^2 + z \ln x, \quad (1, 1, 1).$$

Problem 3. find the derivative of the function at P_0 in the direction of $\mathbf{u}$.

$$f(x, y, z) = xy + yz + zx, \quad P_0(1, -1, 2), \quad \mathbf{u} = 3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}.$$

Problem 4. Sketch the curve $f(x, y) = c$ together with ∇f and the tangent line at the given point. Then write an equation for the tangent line.

$$xy = -4, \quad (2, -2).$$

Problem 5. Let $f(x, y) = \frac{(x-y)}{(x+y)}$. Find the directions $\mathbf{u}$ and the values of $D_{\mathbf{u}} f\left(-\frac{1}{2}, \frac{3}{2}\right)$ for which

a. $D_{\mathbf{u}} f\left(-\frac{1}{2}, \frac{3}{2}\right) = -2$.

b. $D_{\mathbf{u}} f\left(-\frac{1}{2}, \frac{3}{2}\right) = 1$.

Problem 6. In what direction is the derivative of $f(x, y) = xy + y^2$ at $P(3, 2)$ equal to zero?

Problem 7. Find parametric equations for the line tangent to the curve of intersection of the surfaces at the given point.

Surfaces: $x^2 + 2y + 2z = 4, \quad y = 1.$
Point: $(1, 1, 1/2)$.

Problem 8. By about how much will

$$f(x, y, z) = \ln \sqrt{x^2 + y^2 + z^2}$$

change if the point $P(x, y, z)$ moves from $P_0(3, 4, 12)$ a distance of $ds = 0.1$ unit in the direction of $3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$.

In Problems 9–10, find the linearization $L(x, y)$ of the function at each point.

Problem 9. $f(x, y) = 3x - 4y + 5$ at **a.** $(0, 0)$, **b.** $(1, 1)$.

Problem 10. $f(x, y) = e^x \cos y$ at **a.** $(0, 0)$, **b.** $(0, \pi/2)$.

END

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Answers to selected problems

1. $(1, -4)$
2. $(3, 2, -4)$
3. 3
4. $-x + y + 4 = 0$
6. $\pm\big(-\frac{7}{\sqrt{53}}, \frac{2}{\sqrt{53}}\big)$
7. $x = 1 - 2t,\ y = 1,\ z = 1/2 + 2t$
8. 18/910